

Hensel's lemma

MINNETONKA MATH TEAM

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Solving polynomials modulo p^k . Prerequisites: binomial theorem, modular inverses.

1 Theory

We begin with a motivating example:

Example 1.1. Find all x such that $x^2 \equiv 2 \pmod{7^4}$.

Let's try and generalize the previous method. Try to prove the identity by yourself: it looks scary, but it's pretty straightforward by expansion.

Exercise 1.2. Let P be a polynomial with integer coefficients, p^k be a prime power, and $m \leq k$ be a positive integer. Prove the identity

$$P(x + p^k) \equiv P(x) + p^k P'(x) \pmod{p^{k+m}}, \quad (1)$$

where $P'(x)$ is the *derivative* of $P(x)$ (!!).

A corollary is Hensel's lemma, which allows us to “lift” a root r of a polynomial from $\mathbb{Z}/(p^k)$ to a unique root in $\mathbb{Z}/(p^{k+m})$ (where $m \leq k$), as long as $P'(r) \not\equiv 0 \pmod{p}$. Hensel's lemma is most often used when $m = 1$, so we just lift from $\mathbb{Z}/(p^k)$ to $\mathbb{Z}/(p^{k+1})$. I'll introduce this concrete example, which makes the general case clear.

Example 1.3 (A concrete example). Let $P(x) = x^2 - 2$. Use [Equation \(1\)](#) to convince yourself that $P(7^2\lambda + 10)$ hits all multiples of 7^2 modulo 7^3 as λ ranges from 0 to 6, and so in particular there is exactly one value of λ that makes $P(7^2\lambda + 10)$ zero modulo 7^3 .

Theorem 1.4 (Hensel's lemma).

$$P(r + \lambda p^k) \equiv 0 \pmod{p^{k+m}}.$$

Why is the hypothesis $P'(r) \not\equiv 0 \pmod{p}$ necessary?

Note that [Example 1.3](#) was Hensel's lemma for $P(x) = x^2 - 2$, $p^k = 7^2$, $m = 1$, and $r = 10$.

Exercise 1.5. Solve for $s := r + \lambda p^k$ in terms of P and r modulo p^{k+m} . Does it remind you of anything?

Let's talk about how to use Hensel's lemma. Relevant quote:

Your mother can't produce polynomial roots out of thin air, no one can. Roots are the first of the five Principal Exceptions to Gamp's Law of Elemental Transfiguration [...]
It's impossible to make roots out of nothing! You can Summon them if you know where they are, you can transform them, you can lift them if you've already got some...

—*Harry Potter and the Deathly Hallows*

Suppose you want to compute the roots of a polynomial P modulo p^k .

- You must first find the roots of P modulo p : it's impossible to make roots modulo p^k out of nothing!
- For a root r , if $P'(r) \not\equiv 0 \pmod{p}$, we can lift it from $\mathbb{Z}/(p) \rightarrow \mathbb{Z}/(p^2) \rightarrow \mathbb{Z}/(p^4) \rightarrow \cdots \rightarrow \mathbb{Z}/(p^k)$ by repeatedly applying [Exercise 1.5](#).
- For a root r , if $P'(r) \equiv 0 \pmod{p}$, then you should directly apply [Equation \(1\)](#). Then either r has 0 or p lifts modulo p^2 , depending on the value of $P(r)$ modulo p^2 . Repeat to find the lifts of r modulo p^k .

2 Problems

Exercise 2.1 (Brilliant). How many roots does $x^3 - x - 2$ have modulo 2^{10} ?

Exercise 2.2 (AIME I 2024/13). Let p be the least prime number for which there exists an integer n such that $n^4 + 1$ is divisible by p^2 . Find the least positive integer m such that $m^4 + 1$ is divisible by p^2 .

Exercise 2.3 (Brilliant). Find the sum of all prime powers q such that the polynomial $x^3 - 3x - 123$ has at least four roots in $\mathbb{Z}/(q)$.

3 Solutions

Solution to Example 1.1. Motivated by the title of the lecture, we recall the story of Hensel and Gretel. Just as Hensel and Gretel left a trail of breadcrumbs for them to find their way home, we will leave a trail of breadcrumbs ($\sqrt{2}$ modulo 7, 7^2 , and 7^3) to find our way home ($\sqrt{2}$ modulo 7^4).

First, the square roots of 2 modulo 7 are 3 and 4, by guessing.

Now let's try to compute the square roots of 2 modulo 7^2 . Clearly, if $x^2 \equiv 2 \pmod{7^2}$, $x^2 \equiv 2 \pmod{7}$, so $x \equiv 3, 4 \pmod{7}$. WLOG let $x \equiv 3 \pmod{7}$, the computations for $x \equiv 4 \pmod{7}$ are analogous (just take the negative of everything). So let $x = 7\lambda + 3$ for some λ with $0 \leq \lambda < 6$, then we have

$$2 \equiv x^2 = (7\lambda + 3)^2 \equiv (7\lambda)^2 + 2 \cdot 7 \cdot 3 + 3^2 \equiv 7 \cdot 6\lambda + 9 \pmod{7^2}.$$

Note how the $(7\lambda)^2$ term disappears modulo 7^2 , this allows us to solve a linear equation for λ , instead of a quadratic one. Solving the linear equation shows $\lambda \equiv 1 \pmod{7}$, so $\lambda = 1$, giving $x = 10$.

Our final breadcrumb: we now try to compute the square roots of 2 modulo 7^3 ! As before, let $x = 7^2\lambda + 10$ with $0 \leq \lambda < 7$, then we have

$$2 \equiv x^2 = (7^2\lambda + 10)^2 = (7^2\lambda)^2 + 2 \cdot 7^2\lambda \cdot 10 + 10^2 \equiv 7^2 \cdot 20\lambda + 100 \pmod{7^3}.$$

Solving gives $\lambda \equiv 2 \pmod{7}$, so $\lambda = 2$, giving $x = 108$.

Now we find our way home: finally, we compute the square roots of 2 modulo 7^4 ! Again, let $x = 7^3\lambda + 108$ for some $0 \leq \lambda < 7$, then

$$2 \equiv x^2 = (7^3\lambda + 108)^2 \equiv (7^3\lambda)^2 + 2 \cdot 7^3\lambda \cdot 108 + 108^2 \equiv 7^3 \cdot 216\lambda + 108^2 \pmod{7^4}.$$

Solving gives $\lambda \equiv 6 \pmod{7}$, so $\lambda = 6$, giving $x = 2166$. The other square root is $-2166 \equiv 235 \pmod{7^4}$.

Alternatively, we could entirely skip the modulo 7^3 step and go directly from 7^2 to 7^4 : instead, write $x = 7^2\lambda + 10$ for some $0 \leq \lambda < 7^2$. Then

$$2 \equiv x^2 = (7^2\lambda + 10)^2 = (7^2\lambda)^2 + 2 \cdot 7^2\lambda \cdot 10 + 10^2 \equiv 7^2 \cdot 20\lambda + 100 \pmod{7^4}.$$

Solving gives $\lambda = 44$, which gives $x = 2166$ again. ■

Solution to Exercise 1.2. Expand both sides and use the binomial theorem. Any terms containing a p^{2k} , p^{3k} , etc. vanish.

Stupid solution: it's enough to verify the identity when $P(x) = x^n$, since both sides are linear maps of P and the polynomials 1, x , x^2 , etc. form a basis of $\mathbb{Z}[x]$. ■

Solution to Exercise 1.5. From Equation (1),

$$P(r + \lambda p^k) \equiv P(r) + \lambda p^k P'(r) \pmod{p^{k+m}}.$$

We want this to be 0, so solving gives

$$\lambda \equiv -\frac{P(r)}{p^k P'(r)} \pmod{p^m}.$$

Equivalently,

$$r + \lambda p^k \equiv r - \frac{P(r)}{P'(r)} \pmod{p^{k+m}}.$$

This is Newton's method! Not a coincidence, google p -adic numbers. ■

Solution to Exercise 2.1. Let P be the polynomial. Modulo 4, the only root is $2 \pmod{4}$, and since $P'(2) \equiv 1 \pmod{2}$, we may repeatedly apply Hensel's lemma to lift it all the way to $\mathbb{Z}/(2^{10})$. The answer is 1. ■

Solution to Exercise 2.2. For some n , observe that $p^2 \mid n^4 + 1$ implies $n^4 \equiv -1 \pmod{p^2}$ and $n^8 \equiv 1 \pmod{p^2}$, so the order of n modulo p^2 divides 8 but not 4. Therefore, the order of n modulo p^2 is 8, and since $n^{p(p-1)} \equiv 1 \pmod{p^2}$, $8 \mid p(p-1)$, so $p \equiv 1 \pmod{8}$. One may check that if $p \equiv 1 \pmod{8}$, then there is some n such that $p^2 \mid n^4 + 1$, by, say, primitive roots. So we see $p = 17$.

Factoring

$$m^4 + 1 \equiv m^4 - 16 = (m^2 - 4)(m^2 + 4) \equiv (m - 2)(m - 8)(m - 9)(m - 15) \pmod{17},$$

we see $m \equiv 2, 8, 9, 15 \pmod{17}$.

Observe that if $m^4 + 1 \equiv 0 \pmod{17^2}$, then $(-m)^4 + 1 \equiv 0 \pmod{17^2}$ as well. So we only need to consider the residues 2 and 8 modulo 17, and take negatives to get the corresponding ones for 9 and 15. Obviously, $P'(m) = 4m^3 \not\equiv 0 \pmod{17}$, so we may Hensel lift. Applying Exercise 1.5 to lift from modulo $\mathbb{Z}/(17)$ to $\mathbb{Z}/(17^2)$, we see 2 lifts to

$$2 - \frac{2^4 + 1}{4 \cdot 2^3} \equiv 155 \pmod{17^2}$$

and 8 lifts to

$$8 - \frac{8^4 + 1}{4 \cdot 8^3} \equiv 110 \pmod{17^2}.$$

(To simplify the computation, note that we only need to compute the modular inverses of $4 \cdot 2^3$ and $4 \cdot 8^3$ modulo 17, since $17 \mid 2^4 + 1, 8^4 + 1$.) The possible m are $\{155, 110, 289 - 110, 289 - 155\}$, the smallest of which is 110. ■

Solution to Exercise 2.3. Let P be the polynomial and $q = p^k$.

Note that P has at most 3 roots over $\mathbb{Z}/(p)$, since it is a field. Then Hensel's lemma says that unless $P'(r) \equiv P(r) \equiv 0 \pmod{p}$, each root in $\mathbb{Z}/(p^e)$ lifts to a unique root in $\mathbb{Z}/(p^{e+1})$, which eliminates all cases except for those where $P'(r) \equiv P(r) \equiv 0 \pmod{p}$, which gives $p = 5, 11$.

A computation with Equation (1) shows $q = 25, 125, 121$ work, while $q = 625, 1331$ fail. So the answer is $25 + 125 + 121 = 271$. ■